

HDOC - MENU-DRIVEN C PROGRAM

Loops, Switch-Case and Nested Loops

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Problem Statement

Write a menu-driven C program that accepts two positive integers m and n , where $m \leq n$. Use a switch statement and repeatedly display the menu until Exit is selected.

- 1. Print multiplication tables of m and n side by side (1 to 10).
- 2. Display a multiplication grid for integers 1 to m and 1 to n .
- 3. For each integer from m to n , print that integer followed by the next five consecutive integers.
- 4. Exit the program. Display an error message for an invalid menu choice.

Algorithm

1. Start.
2. Read positive integers m and n .
3. If $m > n$, display an error and stop.
4. Display the four menu options inside a loop and read choice.
5. Use switch(choice): option 1 uses one loop from 1 to 10; option 2 uses nested loops; option 3 uses nested loops from m to n ; option 4 exits; default prints an invalid-choice message.
6. Repeat the menu until choice 4 is selected.
7. Stop.

C Program

```
#include <stdio.h>

int main() {
    int m, n, choice, i, j;

    printf("Enter m and n (m <= n): ");
    scanf("%d %d", &m, &n);

    if (m <= 0 || n <= 0 || m > n) {
        printf("Invalid input! Enter positive integers with m <= n.\n");
        return 0;
    }

    do {
        printf("\n1. Multiplication tables of m and n\n");
        printf("2. Multiplication grid\n");
        printf("3. Consecutive integers\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
```

```

switch (choice) {
    case 1:
        printf("\nMultiplication Tables of %d and %d\n", m, n);
        for (i = 1; i <= 10; i++)
            printf("%d x %d = %-4d    %d x %d = %d\n",
                m, i, m*i, n, i, n*i);
        break;

    case 2:
        printf("\nMultiplication Grid (m=%d, n=%d)\n    ", m, n);
        for (j = 1; j <= n; j++) printf("%4d", j);
        printf("\n");
        for (i = 1; i <= m; i++) {
            printf("%2d |", i);
            for (j = 1; j <= n; j++) printf("%4d", i*j);
            printf("\n");
        }
        break;

    case 3:
        printf("\nConsecutive Integers from %d to %d\n", m, n);
        for (i = m; i <= n; i++) {
            printf("%d:", i);
            for (j = i; j <= i + 5; j++) printf(" %d", j);
            printf("\n");
        }
        break;

    case 4:
        printf("Exiting program...\n");
        break;

    default:
        printf("Invalid choice! Please enter 1 to 4.\n");
}
} while (choice != 4);

return 0;
}

```

Test-Case Table

Test	Input	Choice	Expected Result
1	m=3, n=5	1	Tables of 3 and 5 from x1 to x10
2	m=3, n=5	2	3 x 5 multiplication grid
3	m=3, n=5	3	Rows 3 to 5, each with next five integers
4	m=3, n=5	7	Invalid choice message
5	m=3, n=5	4	Program exits

Sample Output (m = 3, n = 5)

```

Enter m and n (m <= n): 3 5

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 1

Multiplication Tables of 3 and 5
3 x 1 = 3    5 x 1 = 5

```

$3 \times 2 = 6$	$5 \times 2 = 10$
$3 \times 3 = 9$	$5 \times 3 = 15$
$3 \times 4 = 12$	$5 \times 4 = 20$
$3 \times 5 = 15$	$5 \times 5 = 25$
$3 \times 6 = 18$	$5 \times 6 = 30$
$3 \times 7 = 21$	$5 \times 7 = 35$
$3 \times 8 = 24$	$5 \times 8 = 40$
$3 \times 9 = 27$	$5 \times 9 = 45$
$3 \times 10 = 30$	$5 \times 10 = 50$

Enter your choice: 2

Multiplication Grid (m=3, n=5)

	1	2	3	4	5
1	1	2	3	4	5
2	2	4	6	8	10
3	3	6	9	12	15

Enter your choice: 3

Consecutive Integers from 3 to 5

3: 3 4 5 6 7 8

4: 4 5 6 7 8 9

5: 5 6 7 8 9 10

Enter your choice: 4

Exiting program...